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Clear cell renal cell carcinoma ontogeny and mechanisms of lethality

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Abstract | The molecular features that define clear cell renal cell carcinoma (ccRCC) initiation and progression are being increasingly defined. The TRACERx Renal studies and others that have described the interaction between tumour genomics and remodelling of the tumour microenvironment provide important new insights into the molecular drivers underlying ccRCC ontogeny and progression. Our understanding of common genomic and chromosomal copy number abnormalities in ccRCC, including chromosome 3p loss, provides a mechanistic framework with which to organize these abnormalities into those that drive tumour initiation events, those that drive tumour progression and those that confer lethality. Truncal mutations in ccRCC, including those in *VHL*, *SET2*, *PBRM1* and *BAP1*, may engender genomic instability and promote defects in DNA repair pathways. The molecular features that arise from these defects enable categorization of ccRCC into clinically and therapeutically relevant subtypes. Consideration of the interaction of these subtypes with the tumour microenvironment reveals that specific mutations seem to modulate immune cell populations in ccRCC tumours. These findings present opportunities for disease prevention, early detection, prognostication and treatment.

Renal cell carcinoma (RCC) affects over 400,000 individuals worldwide per year[1]. It predominantly affects individuals over 60 years of age, and two-thirds of patients are male[1]. Several subvariants of RCC exist, with approximately 70% of individuals being diagnosed with clear cell RCC (ccRCC). Although ccRCC can be successfully treated with surgical or ablative strategies if identified early, up to a third of patients will present with or develop metastases[2]. Metastatic ccRCC is almost uniformly lethal and is biologically distinct from non-metastatic disease.

A number of discoveries have contributed to our understanding of ccRCC ontogeny (FIG. 1). Chromosomal features of three RCC cell lines were described in 1979 (REF.[3]), and chromosome 3p loss was identified in primary tumour samples less than a decade later[4,5]. The discovery of the *VHL* gene in 1993 (REF.[6]) provided the foundation of our current understanding of ccRCC biology, both in sporadic and hereditary settings. Nearly two decades later, use of next-generation sequencing led to the identification of additional mutations associated with ccRCC[7–10]. Additional chromosomal copy number abnormalities were further characterized and found to be associated with disease prognosis[11–15]. Gene expression profiles were subsequently used to subclassify ccRCC into prognostic risk groups[16,17]. In the past few years, studies of genomic heterogeneity have provided further insights into ccRCC clonal evolution[18–21] and raised important questions as to the level of information that is required to adequately characterize the genetic features of any given tumour.

These studies have also raised important questions about the drivers of ccRCC ontogeny and the mechanisms by which ccRCC evolves into a potentially lethal state (FIG. 2). Here, we summarize the consequences of loss of von Hippel–Lindau (VHL) function, the relevance of chromosome 3p deletion and of additional chromosomal events on ccRCC tumour development, and the mechanisms by which these changes might induce the phenotypes observed in ccRCC. We also discuss the potential contribution of drivers of genomic instability, such as defective DNA repair, in ccRCC. This is an area of intense study that has the potential to yield insights into the aetiology and evolution of ccRCC. Finally, we describe ccRCC molecular subtypes and events fostering tumour lethality, as well as the interaction between ccRCC tumour cells and the tumour microenvironment as well as potential therapeutic implications.

Initiating factors in ccRCC ontogeny

Two nearly ubiquitous early events occur in the development of ccRCC: 3p loss and *VHL* mutation or deletion. Together, they result in the functional inactivation of VHL, which drives the early phenotype of ccRCC.

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Key points

- Chromosome 3p loss is an almost universal finding in both hereditary and sporadic clear cell renal cell carcinoma (ccRCC).
- The near ubiquitous loss of a second copy of *VHL* seems to provide a selective advantage for cells, as well as leading to defects in DNA repair and an increase in genomic instability.
- Secondarily mutated genes in ccRCC, including *PBRM1*, *SETD2* and *BAP1*, as well as copy number changes in chromosomes 9p and 14q, are associated with prognostically important molecular and phenotypic characteristics that can be used to classify tumours into subgroups.
- Tumour genomic features are associated with distinct immune phenotypes; for example, *PBRM1* mutations are associated with reduced infiltration of T cells.
- Efforts are underway to link genomic features to specific therapeutic strategies for patients with ccRCC.

Haploinsufficient
A situation in which one copy of a gene is inactivated; for such haploinsufficient genes, gene function is thus altered in heterozygotes, as the remaining (functional) copy does not produce sufficient gene product for normal function.

Von Hippel–Lindau (VHL) disease
A hereditary condition associated with the development of cysts and tumours in multiple organs.

Chromothripsis
A phenomenon by which tens to thousands of chromosomal rearrangements occur; chromothripsis may result from a single catastrophic event during the life history of a cell.

Micronuclei
The small nucleus that forms when a chromosome or a fragment of a chromosome is not incorporated into one of the daughter nuclei during mitosis. It is usually a sign of genotoxic events and chromosomal instability.

Aneugens
Substances that cause a daughter cell to have an abnormal number of chromosomes.

Clastogens
Mutagenic agents that give rise to or induce disruption or breakages of chromosomes.

Chromosome bridges
Attachments of the telomeric ends of sister chromatids during mitosis, specifically during anaphase, preventing segregation into respective daughter cells.

Chromosome 3p deletion

ccRCC is characterized by a near-universal loss of most or all of chromosome 3p[10,13,22]. A number of genes, including *PBRM1*, *SETD2* and *BAP1* (encoding polybromo 1 (PBRM1), SET domain containing 2 (SETD2) and BRCA-associated protein 1, respectively) reside on chromosome 3p, and these genes are mutated at relatively high frequencies in ccRCC[7,9,10]. Additionally, several genes on chromosome 3p may be haploinsufficient, including the mismatch repair gene *MLH1* (REF.[23]) and *SETD2* (REF.[24]).

Chromosome 3p loss was described as early as 1987 (REF.[4]), and the timing of 3p loss relative to subsequent genomic alterations was defined in 2000 (REF.[15]). More recently, the TRACERx Renal programme, which prospectively collected tumour tissue samples from over 100 patients with ccRCC including multiple samples from each tumor[19–21], showed that loss of 3p is the earliest event in RCC ontogeny, likely occurring in adolescence[19,20]. A study of patients with Von Hippel–Lindau (VHL) disease undergoing multiple ccRCC resections demonstrated that loss of chromosome 3p in synchronously arising tumours occurred via different 3p breakpoint locations, or through whole chromosome 3 loss[25]. These findings suggest that although 3p loss itself is a necessary event for carcinogenesis, the process leading to that loss is not driven by a specific chromosomal breakpoint.

The TRACERx Renal study also demonstrated that chromothripsis is a likely driver of chromosome 3p loss, in association with chromosome 5q gain[19,20]. Chromothripsis might result from the formation of micronuclei during mitosis, which induce a caustic environment for the DNA contained within and drive DNA damage, mutation and chromothripsis[26]. Underlying causes of micronuclei formation include aneugens and clastogens. Clastogens are generally associated with the formation of acentric chromosome fragments[27], which lack a centromere and fail to attach to the mitotic spindle, thereby inducing the formation of micronuclei. The presence of chromosome bridges during telophase can also lead to DNA breakage during cytokinesis and induce the formation of micronuclei[28]. Hypoxia and oxidative stress — conditions that are potentially present in the kidney proximal tubule — have also been linked to micronucleus formation[27]. Other physical changes in the cell, for example, due to alterations in pressure and/or temperature, or exposure to radiation, UV or ultrasound can induce micronuclei formation[27]. Of note, the kidney proximal tubule and nephron represents a fairly harsh environment, characterized by intense gradients of oxygen, salts and nutrients. Thus, conditions around the RCC cell of origin may create a 'perfect storm' that permits the development of micronuclei, chromothripsis and 3p loss.

VHL loss

Loss or mutation of the *VHL* gene is generally considered to be one of the obligate initiating steps in the development of ccRCC. Germline loss of *VHL* is inherited in an autosomal-dominant manner in families with VHL disease[6], and these individuals often develop multifocal, bilateral ccRCC[29,30]. Somatic loss of *VHL* is observed in the majority of patients with sporadic ccRCC[31] and, based on findings from the TRACERx studies, may follow loss of chromosome 3p in these patients[19].

VHL and HIF regulation. The VHL protein is an E3 ubiquitin ligase, best known for its role in the ubiquitylation of the prolyl hydroxylated transcription factors hypoxia-inducible factor 1α (HIF1α) and HIF2α[32–34] under normoxic conditions to induce their proteolytic degradation[35]. HIF1α and HIF2α regulate the transcription of a number of genes involved in angiogenesis, metabolism and chromatin remodeling[36]. Even before the functional link between VHL and HIF was known, ccRCC was demonstrated to be associated with increased levels of *VEGF* (encoding vascular endothelial growth factor) and *PDGF* (encoding platelet-derived growth factor)[37]. Moreover, HIF-mediated transcription of pro-angiogenic factors including *VEGF* was shown to facilitate neovasculogenesis in human tumour cell mouse xenograft models[33].

HIF1α and HIF2α regulate common gene sets, as well as genes that are uniquely regulated by HIF1α and HIF2α[38–40]. HIF1α regulates a number of genes involved in glucose uptake and glycolysis, such as *PKP1* and *LDHA* (which encode pyruvate dehydrogenase kinase[41] and lactate dehydrogenase A[42], respectively) as well as those involved in mitochondrial autophagy such as *BNIP3* (REF.[43]) and *BNIP3L*[44]. Expression of *AXL*[45] — a member of the tumour-associated macrophage (TAM) family of regulatory proteins[46,47] — is also driven by HIF1α[48]. *AXL* levels are associated with a poor prognosis in ccRCC[46,48] and treatment with small molecule VEGFR inhibitor sunitinib increases levels of AXL[49].The induction of AXL by sunitinib might underlie the development of resistance to antiangiogenic therapy that is frequently observed in response to this agent. HIF2α regulates *EPO, CCND1* and *TGFA* (which encode erythropoietin, cyclin D1 and transforming growth factor-α, respectively)[36,39]. Some evidence suggests that HIF2α, rather than HIF1α, is the primary driver of ccRCC oncogenesis, with HIF2α being both necessary and sufficient for the growth of VHL-null cell lines[36,50]. However, other studies in animal models have implicated HIF1α[51] rather than HIF2α[52,53] as the primary driver of renal carcinogenesis. Thus, questions remain as to the exact role of specific HIF isoforms in the development of ccRCC, perhaps implying

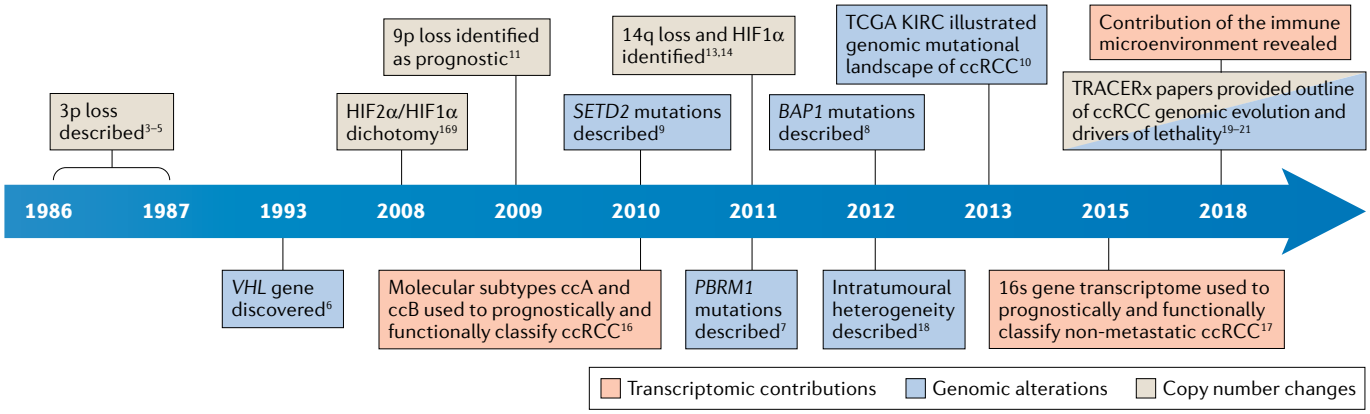

Fig. 1 | **Key discoveries in ccRCC genomics.** A number of discoveries have contributed to our current understanding of clear cell renal cell carcinoma (ccRCC) ontogeny. KIRC, kidney renal clear cell carcinoma; TCGA, The Cancer Genome Atlas.

a temporal process whereby one factor plays a larger part than the other at various stages of disease. A 2016 study showed that loss of either HIF1α or HIF2α abrogated tumour formation in mice lacking *Vhl* and *Trp53* (REF.[54]), indicating that both HIF isoforms are required for tumour initiation in that system.

VHL and non-HIF targets. Beyond HIF, VHL regulates a number of targets, pathways and processes[55], including *AKT*[56], *NFKB*[57–59] and *TBK1* (REF.[60]), as well as components of the extracellular matrix (ECM), the primary cilium and factors involved in mitosis (FIG. 3). VHL is also linked to the regulation of DNA repair[61–64].

Loss of VHL is associated with an upregulation of NF-κB activity[57] through HIF-dependent induction of TGFα. However, VHL also acts as an adaptor protein to promote the inhibitory phosphorylation of the NF-κB agonist CARD9 by casein kinase 2 (CK2); thus, the loss of VHL in ccRCC may release this inhibition and induce NF-κB activation through HIF-independent mechanisms[58]. Moreover, ZHX2, an activator of NF-κB, is directly targeted and downregulated by VHL through ubiquitylation and proteasomal degradation[65]. Thus, the loss of VHL leads to coordinated stabilization of factors that act independently to promote NF-κB signalling (FIG. 3a).

Several lines of evidence also suggest that VHL is involved in the regulation of intercellular junctions through the maintenance of adherens junction components[66] and ECM formation[67–70] (FIG. 3b). Binding of VHL to fibronectin is necessary for correct ECM assembly around cells[68]. Of note, all tumour-causing *VHL* mutants have been reported to be defective in in fibronectin binding[71–73]. The binding of VHL to fibronectin is regulated by CK2 and requires CK2-mediated phosphorylation of VHL at Ser33, Ser38 and Ser43 (REF.[74]). Mutation of these three serines was associated with eventual ccRCC tumour growth in a xenograft model[74]. Conversely, inhibition of CK2 resulted in VHL stabilization owing to decreased N-terminal cleavage and proteasomal degradation, suggesting that regulation by CK2 is linked to VHL protein stability[75]. VHL also binds to collagen IV and this interaction is necessary for collagen IV assembly in the ECM[69]. Loss of VHL induces dysregulation of collagen IV homeostasis in vitro and in vivo[69].

VHL also enhances microtubule stability[76] — an interesting observation given the link between SETD2 and methylation of cytoskeletal components (see below). This particular role has implications for several important microtubule structures, including the primary cilium and mitotic spindle. The central core of the ciliary axoneme comprises microtubules, and the loss of VHL results in defective ciliogenesis[77–79]. As a result, VHL disease is often classified as a 'ciliopathy', along with other diseases linked to defects of the primary cilium such as polycystic kidney disease[80]. Aurora kinase A (AURKA) is a kinase that inhibits ciliogenesis and reduces cilia stability[81]. VHL monoubiquitinates rather than polyubiquitylates AURKA in a prolyl hydroxylase-independent manner, thereby targeting AURKA for degradation. This degradation process is required to maintain the primary cilium in quiescent cells, but is absent following the loss of VHL, leading to AURKA activation and likely contributing to dysregulation of ciliogenesis and cilia instability[81].

VHL has also been linked to mitotic spindle function[82–84]. A 2009 study showed that loss of VHL resulted in reduced levels of the mitotic spindle assembly checkpoint protein MAD2, which was associated with spindle misorientation and loss of normal mitotic checkpoint control in an isoform-specific manner[82]. A further study demonstrated that translation of MAD2 is regulated by the microRNA miR-28-5p in a VHL-dependent manner; loss of VHL resulted in increased levels of miR-28-5p and decreased translation of MAD2 (REF.[84]). Furthermore, induction of ischaemic injury in mice with kidney-specific deletion of *Vhl* led to spindle misorientation and aneuploidy immediately after injury. Longer term follow-up revealed the development of cysts, ccRCC-type cells and dysplasia[83]. Thus, VHL might have a role in the regulation of both mitotic and non-mitotic cells via maintenance of microtubular homeostasis (FIG. 3c).

VHL also downregulates AKT[56], a key component of the PI3K pathway. Upregulation of the PI3K pathway has been associated with aggressive behaviour in many cancers[85], including ccRCC[86,87]. Increased PI3K

Cytokinesis
The physical process of cell division, which divides the cytoplasm of a parental cell into two daughter cells.

Aneuploidy
An abnormal number of chromosomes in a cell.

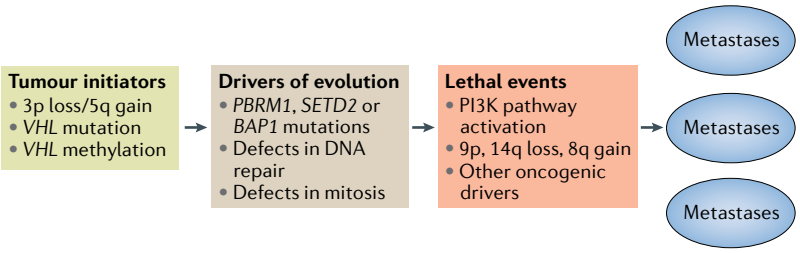

Fig. 2 | **Key events in clear cell renal cell carcinoma progression.** Tumour initiators include chromosome 3p loss and mutations in *VHL*. Tumour evolution is driven by DNA repair defects and errors in mitosis that create additional aneuploidy. The acquisition of key chromosomal gains and losses, mutations in PI3K pathway elements and other oncogenic driver mutations confer lethal potential to intratumoural clones and enhance the probability of metastasis. Of note, interactions exist between factors involved in tumour evolution and lethality; for example, *BAP1* mutations, which are a driver of tumour evolution, are sufficient to create a tumour of considerable lethality.

signalling is associated with increased cell division as well as with enhanced survival, and decreased autophagy, all of which favour tumour growth[88]. Increased phosphorylation of AKT is associated with increased tumour aggressivity and with cytoplasmic translocation and deactivation of the cell cycle checkpoint p27 (REF.[89]).

VHL mutations and development of animal models for ccRCC. The observation that germline *VHL* mutations in humans drive the development of pleiotropic lesions, including ccRCC, led to the assumption that the loss of *VHL* is sufficient for tumorigenesis. However, a number of studies have shown that knockout or mutation of *Vhl* does not engender the development of ccRCC in mouse models[90–92], although the homozygous knockout of *Vhl* is embryonically lethal[93]. The first global heterozygous *Vhl*-knockout mouse described in 2001 did not produce an RCC phenotype[90]. Subsequent efforts to create kidney-specific biallelic *Vhl* knockouts were also unable to produce bona fide RCC lesions, despite the use of various kidney-specific promoters[94], including a proximal tubule-specific *Pepck*–Cre[95], a pan-tubular *Pax8*–Cre[53,96] and a thick ascending limb-specific *Thp*–Cre[97]. Moreover, studies of renal cysts from patients with VHL disease have shown that although both copies of *VHL* are inactivated in these benign lesions[98], other characteristic hallmarks of ccRCC are often not present[99]. These findings indicate that although *VHL* loss is necessary for ccRCC development, it is not necessarily sufficient.

The lack of success with single-gene knockout models led to the combined knockout of *Vhl* with inactivation of other potential RCC driver genes, many of which used the proximal tubule-specific *Ksp1.3* to drive *Cre* recombinase and inactivate paired or multiple genes in a kidney-specific manner[100,101]. Combined inactivation of *Pten* and *Vhl* generated cysts in mice[101], whereas the combined inactivation of *Trp53* and *Vhl* generated simple cysts, atypical cysts and neoplasms with activation of mTORC1 and elevated expression of MYC[100]. However, neither of these models developed tumours with metastatic potential. Combined *Ksp1.3*–Cre driven inactivation of *Trp53, Vhl* and *Rb* produced neoplastic lesions with histological and transcriptomic features that resembled those of human ccRCC, but still did not demonstrate metastatic potential[102].

More recent efforts to create models of ccRCC have used polygenic approaches or focused on mutations in chromatin remodelling genes. These approaches have produced models that more closely resemble human ccRCC[103–105], as described later.

Additional mutations in ccRCC

Several genes other than *VHL*, all of which are found on chromosome 3p, are also mutated at a relatively high frequency in ccRCC[7–9]. All of these genes are involved in chromatin remodelling and are likely associated with both the convergent and divergent phenotypic characteristics of this disease[106].

PBRM1. *PBRM1* is the most commonly mutated gene after *VHL*. Human *PBRM1* generates a 1,689-amino acid protein, that comprises six bromodomains, two bromo-associated homology domains and a C-terminal high-mobility group[107]. PBRM1 is a key component of PBAF — an ATP-dependent chromatin remodelling complex of the switch/sucrose non-fermenting (SWI/SNF) family[108] that governs nucleosome structure, nucleosome positioning and gene transcription. Loss of *PBRM1* and therefore PBAF activity is likely to considerably disturb the highly organized structure of chromatin and disrupt transcription. Numerous studies have identified *PBRM1* mutations in ccRCC[7,18,109], with an incidence of 30–40%, although the most aggressive sarcomatoid variants tend to lack these mutations[110]. *PBRM1* mutations precede the development of *SETD2* mutations, suggesting that this sequence of mutations might contribute to subclonal tumour evolution[19].

Several studies have provided mechanistic insights into the consequences of *PBRM1* and PBAF loss in ccRCC. A 2017 study demonstrated that *PBRM1* knockdown is associated with increased cellular growth and a HIF transcriptional signature[111]. Similarly, *Pbrm1* deletion enhanced the upregulation of HIF1α and STAT3 induced by loss of *Vhl* and increased mTOR signalling, thereby enabling the development of kidney lesions in transgenic mice[104]. A further study showed that *PBRM1* and other chromatin remodelling genes that are lost in ccRCC normally act to antagonize tumour growth in a process dependent on the interferon-responsive transcriptional regulator *ISGF3*; loss of *PBRM1* relieves this inhibition, leading to tumour growth[112]. In mice, knockout of *Pbrm1* rescued *Vhl* loss-induced replication stress and enhanced tumour growth[63]. In another mouse study, combined inactivation of *Pbrm1* and *Vhl* under a *Pax8*–Cre driver induced a low-grade cancer phenotype that partly recapitulated the human phenotype[103]. Loss of *PBRM1* has also been shown to downregulate interferon-γ (IFNγ)-mediated signalling, decreasing immunogenicity[113]. PBRM1 also acts as an acetyl-lysine reader for p53, and thereby regulates p53 transcriptional activity as demonstrated by the finding that loss of *PBRM1* decreases p53-dependent transcriptional regulation of *CDKN1A* (which encodes p21)[114]. Similar to VHL, PBRM1 also has a role in genomic stability.

Replication stress
Any condition that compromises the faithful duplication of the genome once per cell cycle.

Acetyl-lysine reader
A protein that recognizes acetylated lysine residues in promoters and enhancers and has an active role in gene transcription.

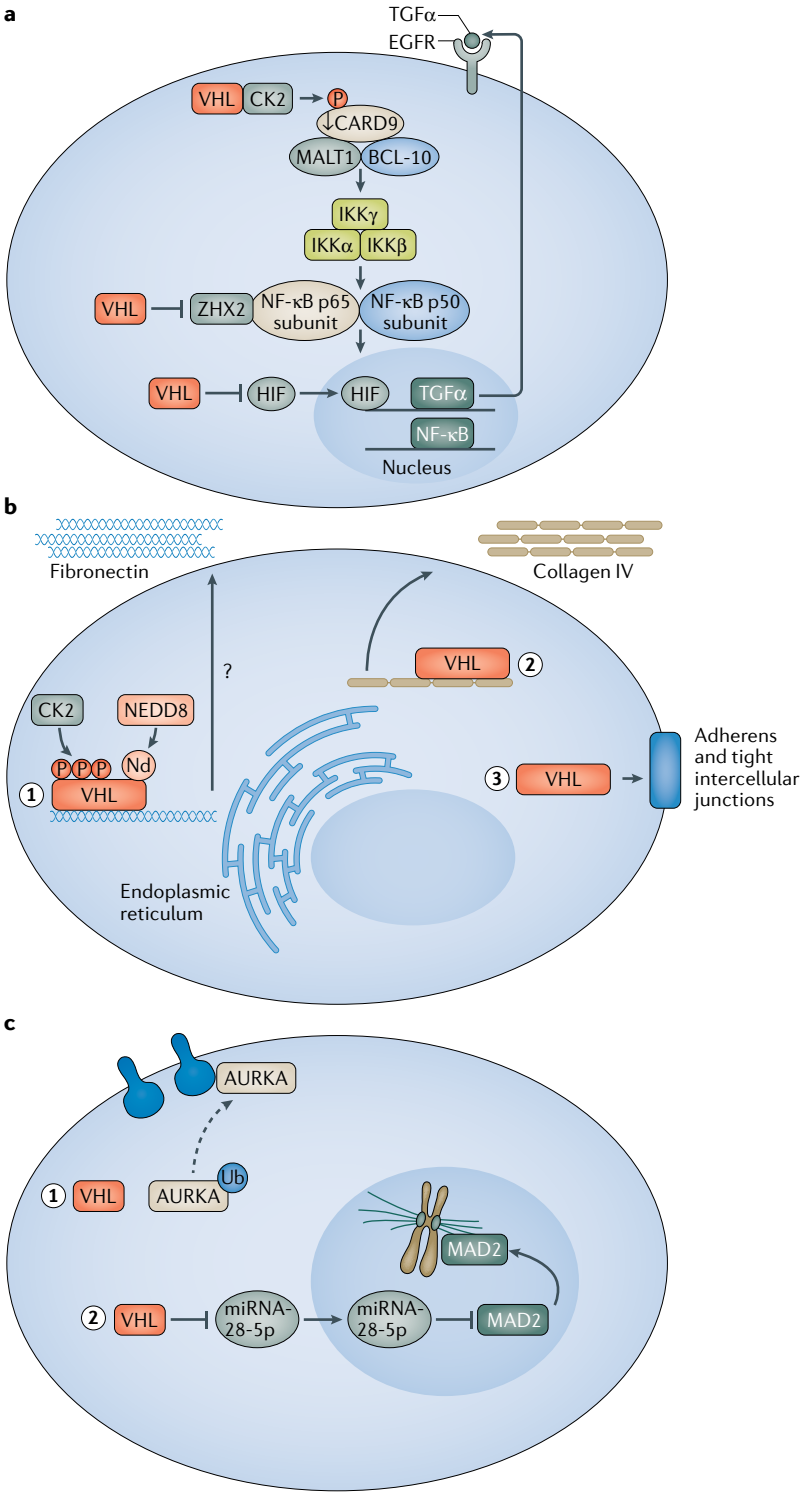

Fig. 3 | **Non-HIF targets of VHL. a** | von Hippel–Lindau (VHL) regulates NF-κB through hypoxia-inducible factor (HIF)-dependent and independent mechanisms. Loss of *VHL* increases transcription of *TGFA*, which induces a cell autonomous increase in NF-κB via activation of the TGFα receptor EGFR and inhibition of IKK. However, CK2-mediated inhibition of the NF-κB agonist CARD9, along with cofactors Malt-1 and Bcl10, is also enhanced by VHL, whereas the NF-κB activator, ZHX2, is normally inhibited by VHL, decreasing ZHX2-mediated nuclear localization of the p65 subunit of NF-κB. **b** | Extracellular matrix homeostasis is dependent on the interaction between VHL and fibronectin and requires CK2-mediated phosphorylation and neddylation of VHL (1). Loss of VHL therefore results in defective or deficient extracellular matrix. VHL is involved in the maintenence of collagen IV homeostasis through unclear mechanisms (2). Finally, VHL regulates intercellular and adherens junctions to maintain cell–cell interactions (3). **c** | VHL regulates the primary cilium and mitosis. VHL maintains primary cilium homeostasis by targeting AURKA for degradation, thereby preventing AURKA-mediated cilia disassembly (1). VHL also supports mitotic function by enhancing the function of the mitotic spindle assembly checkpoint protein MAD2 through downregulation of the MAD2 inhibitory miRNA-28-5p (2).

It localizes to the kinetochore[115] and in yeast, the *PBRM1* homologue *Rsc2* is essential for chromosome arm cohesions[116,117]; inhibition of PBAF activity resulted in genomic instability and aneuploidy in human cell lines[118]. Thus, the potential impact of loss of *PBRM1* includes the augmentation of hypoxia signalling, modulation of replication stress, impairment of p53-mediated cell cycle regulation, inhibition of the interferon response and genomic instability. However, in ccRCC, in the absence of additional mutations in *SETD2*, these changes seem to produce tumours that are not overly aggressive and are immunologically cold[113,119,120].

Kinetochore
A large protein complex that assembles at the centromere of a chromosome and functions to connect the chromosome to microtubules in the mitotic spindle.

BAP1

Another chromatin remodeller that is located on chromosome 3p and inactivated in ccRCC is the BAP1 deubiquitinase[121]. The discovery of *BAP1* mutations in ccRCC arose from the study of patient-derived xenograft models[8]. The *BAP1* gene produces a 729-amino acid protein that contains several functional domains: a ubiquitin C-terminal hydrolase (UCH) domain, a BRCA1-associated RING domain protein 1 (BARD1)-binding domain, an HCF-C1 binding motif (HBM) and a nuclear localization signal[8]. *BAP1* is mutated in 10–20% of ccRCCs[8,122]. Lysine 119 on histone 2a (H2AK119) — which is involved in polycomb complex-mediated gene repression — is a target of BAP1 (REFS[8,123]), which normally counteracts polycomb PRC1-mediated H2AK119 monoubiquitination and consequent compaction of chromatin structures to repress gene transcription at target loci[124]. Studies in mice have shown that BAP1 also interacts with the cell cycle regulator host cell factor-1 (HCF-1), the signal transduction modulator *O*-linked *N*-acetylglucosamine transferase (OGT) and the polycomb group proteins ASXL1 and ASXL2. Levels of OGT and HCF-1 were decreased by deletion of *Bap1*, and were associated with myeloid transformation[125]. A subsequent study reported that BAP1 modulated gene expression in a cell-type specific manner, and that loss of BAP1 promoted tumorigenesis in cells that did not engage an RNF2-dependent apoptotic programme[126].

Similar to both VHL and PBRM1, BAP1 also has a role in the maintenance of genomic stability. BAP1 associates with the mitotic spindle regulating microspherule protein 1 (MCRS1), and loss of BAP1 led to increased chromosomal instability[127].

In mice, heterozygous deletion of *Bap1* coupled with homozygous loss of *Vhl* using a *Six2*–Cre driver[128] (*Six2*–Cre;*Vhl*$^{F/F}$;*Bap1*$^{F/+}$) to drive kidney-specific gene inactivation produces kidney neoplasms[129]. Combined heterozygous inactivation of *Bap1* and *Vhl* under a *Pax8*–Cre driver (*Pax8*–Cre;*Vhl*$^{F/F}$;*Bap1*$^{F/+}$ mice) also produced neoplastic kidney lesions with nuclear pleomorphisms, nucleolar prominence, atypia, mitosis and elevated phospho-S6 expression, which, although indicative of aberrant cellular behaviour, does not fully phenocopy a typical ccRCC. By comparison, heterozygous deletion of *Pbrm1* together with *Vhl* (*Pax8*–Cre;*Vhl*$^{F/F}$;*Pbrm1*$^{F/+}$) failed to induce kidney lesions and, as mentioned above, homozygous deletion of *Pbrm1* together with *Vhl* (*Pax8*–Cre;*Vhl*$^{F/F}$;*Pbrm1*$^{F/F}$) produced low-grade tumours with only moderate levels of cytoplasmic lipids and glycogen, a relative lack of nuclear atypia, and an absence of phospho-S6 staining[103]. Loss of *BAP1* is strongly associated with a poor prognosis among patients with ccRCC, even in individuals with clinically low-stage tumours[130]. In patients treated with targeted therapy such as antiangiogenic therapy, the presence of mutations in *BAP1* was associated with worse outcomes than those associated with mutations in *PBRM1* (REF.[131]). Germline mutations in *BAP1* have also been identified in patients with familial ccRCC[132]. In the TRACERx Renal study, *BAP1* mutations were associated with a high weighted genome instability index (wGII), low intratumoural heterogeneity (ITH) tumours, suggesting that the driver function of *BAP1* loss is sufficient to provide ccRCC clones with a clear fitness advantage and lethal potential[19].

SETD2

SETD2 is another chromatin remodelling gene and a tumour suppressor in ccRCC[9]. *SETD2* mutations were first discovered through early high-throughput genomic sequencing efforts[9] and were subsequently validated by findings from the Cancer Genome Atlas (TCGA) project[10]. *SETD2* is mutated in 10–20% of cases of ccRCC[9,10] Its encoded protein, SETD2, is a 2,564-amino acid non-redundant histone H3 lysine 36 methyltransferase, which contains a catalytic SET domain and a C-terminal Set2–Rpb1 interacting (SRI) domain, and specifically marks the nucleosomes of actively transcribed genes with H3K36me3. In yeast, this mark, along with demethylation (H3K36me2), has a repressive role and prevents initiation of cryptic transcription from alternate internal transcriptional start sites of genes, which can result in the expression of alternatively spliced genes that may possess oncogenic potential[133]. However, analysis of selective *Set2* mutants that isolate monomethylation, dimethylation and trimethylation states suggests that the isolated loss of trimethylation, as occurs in mammalian cells with *SETD2* loss, is not sufficient to promote a cryptic initiation phenotype[134]. Instead, the chromatin effects of this mark seem to have important roles in the regulation of DNA repair[135].

Beyond DNA repair, SETD2 contributes to genomic stability by promoting the methylation of microtubules[136,137]. SETD2 trimethylates mitotic spindle α-tubulin on lysine 40 (αTubK40me3)[136]. SETD2-deficient cells demonstrate defects in the mitotic spindle and in cytokinesis, with increased formation of chromosome bridges and micronuclei[136]. Mutation of the SET and SRI domains impaired the ability of SETD2 to methylate microtubules, resulting in the formation of more chromosome bridges and lagging chromosomes than observed with wild-type *SETD2* (REF.[136]). These findings indicate that in addition to its catalytic SET domain, a functional SRI domain is also required for αTubK40me3 (REF.[136]). Monoallelic *SETD2* deletion, as occurs in cells with loss of chromosome 3p, abrogates trimethylation of spindle microtubules on αTubK40, but not methylation of histones on H3K36; however, this effect is sufficient to induce genomic instability[24]. It is possible that monoallelic loss of *SETD2* might also contribute to the steady increase in chromosomal copy number abnormalities that is seen as tumours increase in size and stage[11,13]. New findings show that in addition to trimethylating α-tubulin, SETD2 also methylates cytoskeletal actin, promotes polymerization of actin filaments and enhances cell motility[138]. As noted above, *SETD2* mutations can arise independently of, or occur in conjunction with, *PBRM1* mutations in ccRCC[19]. Interestingly, functional convergence between SETD2 and PBRM1 was revealed with the discovery that PBRM1 recognizes the αTubK40me3 SETD2 mark on microtubules[139]. In that study, PBRM1 localized the PBAF complex to the mitotic spindle to maintain genomic stability. As mentioned earlier and consistent with a reader–writer relationship between PBRM1 and SETD2, findings from the TRACERx Renal study show that loss of *PBRM1* precedes loss of *SETD2* in patients in whom mutations in these two genes co-occur. Currently, no published animal tumour model of *SETD2* inactivation in ccRCC exists.

Aetiology of DNA repair defects

As described above, the initiating events for ccRCC development involve loss of biallelic *VHL* and chromosome 3p[10,19,31]. However, ccRCCs also demonstrate a relatively low number of mutations per megabase, with a fairly low variability[10,20] and a stage-dependent increase in the number of chromosomal copy number alterations[13]. The mechanisms that drive these genomic events are beginning to be elucidated, and have potential to provide new therapeutic approaches in ccRCC.

Mismatch repair defects

Several pathways contribute to the repair of specific types of DNA damage to maintain genome stability; defects in the DNA damage response can induce genomic instability in cells and aid the initiation of cancer, including ccRCC[140]. A 2013 analysis of mutational patterns in tumours with known genomic defects, using an algorithm that considered nucleotides surrounding mutational sites led to the identification of 21 different patterns that were associated with known germline deficiencies[141]. Using this approach, ccRCC was linked to a pattern associated with ageing, as well as a pattern associated with DNA mismatch repair in a subset of individuals. Several potential mechanisms exist through which a deficiency in mismatch repair could lead to ccRCC (FIG. 4). *MLH1* is a gene that encodes MutL homologue

Genome instability index
An estimate of the proportion of the genome with aberrant copy number compared with the median ploidy, weighted on a per chromosome basis.

Intratumoural heterogeneity
(ITH). The presence of distinct tumour cell populations within the same tumour specimen.

Cryptic transcription
Transcripts initiated from intragenic promoters that are usually not accessible to transcriptional machinery.

DNA mismatch repair
A process for recognizing and repairing erroneous insertion, deletion and misincorporation of bases that can arise during DNA replication and recombination, or through DNA damage.

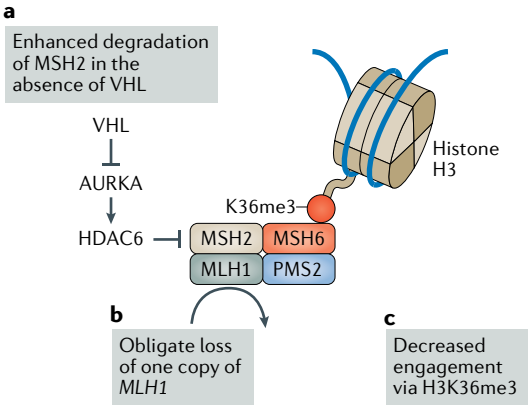

Fig. 4 | **Factors influencing mismatch repair in ccRCC.** Multiple mismatch repair mechanisms are affected in clear cell renal cell carcinoma (ccRCC). **a** | Loss of VHL itself alters HDAC6 regulation via AURKA, effectively suppressing the DNA mismatch repair protein MSH2 via protein degradation. **b** | The mismatch repair gene *MLH1* may exhibit haploinsufficiency as a result of chromosome 3p deletion. **c** | Loss of MSH6 targeting H3K36me3 in the absence of *SETD2* can result in the uncoupling of transcription-coupled repair mechanisms and impair the efficient targeting of DNA repair mechanisms to essential segments of the genome.

1 — an essential component of the mismatch repair complex hMutLα, which consists of MLH1 and PMS2 proteins — and is found on chromosome 3p. Evidence for *MLH1* haploinsufficiency has been reported in pancreatic cancer[23]. Loss of *VHL* also upregulates levels of histone deacetylase 6 (HDAC6)[142], which results in the deacetylation and ubiquitylation of the DNA mismatch repair protein MutS protein homologue 2 (MSH2), inducing its degradation[143]. HDAC6-induced downregulation of MSH2 also reduces therapeutic sensitivity to DNA-damaging agents and cellular DNA mismatch repair activities[143] In addition, SETD2-mediated trimethylation of H3K36 provides an essential docking site for the hMutSα–hMutLα complex[144]. Specifically, the hMutSα complex, which consists of MSH2 and MSH6, is directed to actively transcribed genes through interaction with the H3K36me3 mark. Thus, despite the fact that mismatch repair components remain intact in the absence of H3K36me3, DNA repair might be less efficiently targeted to the most relevant areas of the genome. This interaction between SETD2-mediated trimethylation of H3K36 and the hMutsα–hMutlα complex implies the critical activity of H3K36me3 in the maintenance of genome stability, by targeting mismatch repair to the active coding regions.

Together, these data point to multiple potential defects in the integrity of mismatch repair in ccRCC. Of note, the rate of microsatellite instability seen in ccRCC is very low, suggesting that mismatch repair is intact[145,146]. In addition, ccRCC has a relatively low absolute mutation rate, with frequencies approximately one-tenth those seen in tumours from patients with Lynch syndrome[141]. These findings suggest that a non-classical form of mismatch repair defect may be operative in ccRCC that induces a lower rate of mismatch repair errors than seen in Lynch syndrome. Further study is needed to clearly elucidate the role of mismatch repair defects in ccRCC mutagenesis.

Homologous recombination repair defects

The two main pathways involved in double-stranded DNA repair include non-homologous end joining (NHEJ), which acts throughout the cell cycle, and homologous recombination repair (HRR), which is operative in the S phase and requires two DNA strands for error-free repair[147]. NHEJ tends to be more prone to error than HRR, and defects in HRR are associated with increased mutational frequency and cancer predisposition, classically manifested in families bearing mutations in HRR-related genes, such as *BRCA1* (REFS[148,149]), *BRCA2* (REF.[150]), *PALB2* (REF.[151]) and *ATM*[152].

A number of groups have assessed the integrity of HRR in ccRCCs (FIG. 5). One study reported that SOCS2 mediates monoubiquitination of VHL at lysine 63 (REF.[61]) and that loss of VHL, or *VHL* mutations that compromise its ubiquitylation at lysine 63 attenuated the DNA-damage response, resulting in decreased HRR and persistence of double-stranded DNA breaks[61]. The finding that loss of *VHL* is associated with an HRR defect is supported by other studies[62]. Loss of *Vhl* also enhanced replication stress in mice[63], suggesting a role for VHL in the maintenance of genomic integrity during mitosis. A link between hypoxia and HRR defects has also been demonstrated, which is driven mainly by the transcriptional regulation of key HRR components[153–155]. VHL also associates with and decreases the activity of KAT5 (also known as TIP60) — an acetyltransferase that is associated with p53 activation[156]. TIP60 is essential for H4K16 acetylation, which evicts p53-binding protein 1 (53BP1) from chromatin and favours the use of HRR as opposed to NHEJ in DNA repair[157]. TIP60 is also needed for ATM acetylation[158], which is necessary for ATM autophosphorylation and activation. A 2018 study extended this observation to demonstrate that the VHL- and TIP60-mediated shift in DNA damage repair from error-prone NHEJ to HRR operates in a HIF-independent manner, and that ATM activation is impaired in VHL-deficient cells owing to decreased ATM acetylation[64].

SETD2 is also involved in regulating a number of processes associated with HRR. The best characterized of these is the recruitment of DNA repair machinery to the sites of actively transcribed genes[159,160]. The preferential placement of H3K36me3 at sites of active transcription by SETD2 provides a convenient mechanism for 'readers' involved in DNA repair to be preferentially recruited to high priority genomic sites for repair[161]. Loss of SETD2 specifically alters the efficiency of repair at these active sites[160]. However, further work has demonstrated that SETD2 also has an active role in mediating DNA repair by engaging the p53-dependent cell cycle checkpoint; SETD2-knockout cells show inactivation of this p53-mediated checkpoint, as shown by a failure to increase total p53 levels and p53 phosphorylation, which prevents cells from halting cell cycle progression and executing effective HRR[160]. This defect in repair is not unlike that observed with mutations in the DNA repair gene *RAD51*, which results in a signature of

Microsatellite instability
Genetic predisposition to mutation that results from impaired DNA mismatch repair.

Lynch syndrome
An inherited disorder that arises from germline genetic defects in mismatch repair components and increases the risk of many types of cancer, particularly cancers of the colon.

Non-homologous end joining
(NHEJ). A pathway that repairs double-strand breaks in DNA without the need for a homologous template.

Homologous recombination repair
(HRR). A DNA repair process in which an intact DNA molecule is used as a template for the repair process, in particular during the S phase.

Cell cycle checkpoint
Control mechanisms in the eukaryotic cell cycle that ensure its proper progression.

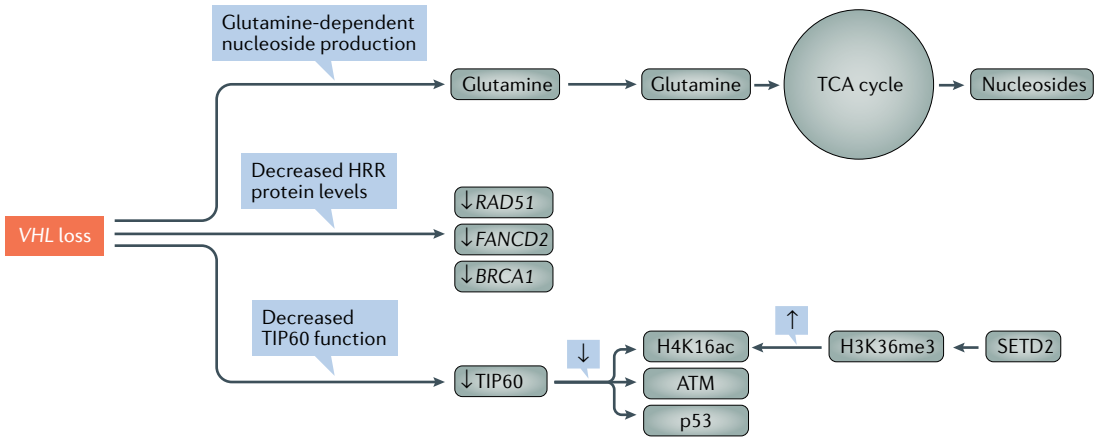

Fig. 5 | **Factors influencing homologous recombination repair in clear cell renal cell carcinoma. a** | Loss of *VHL* induces a metabolic change that creates glutamine dependence for nucleoside production. This change increases the risk of a nucleoside shortfall, which can create vulnerabilities during mitosis-dependent DNA repair. **b** | VHL loss affects key genes involved in homologous recombination repair through an HIF-dependent mechanism. **c** | VHL interacts with KAT5 (also known as Tip60), which acetylates and regulates ATM, p53 and H4K16. TIP60-dependent ATM acetylation is necessary for full ATM activation, as is p53 acetylation. SETD2-dependent H3K36 trimethylation enhances H4K16 acetylation. HRR, homologous recombination repair; TCA, tricarboxylic acid.

DNA defects, one hallmark of which is microdeletions due to a reliance on microhomology repair mechanisms, and a reliance in the absence of SETD2 on NHEJ[159].

The effect of impaired HRR on replication stress and the accumulation of further mutations has also been considered a source of heterogeneity and tumour evolution in ccRCC[162]. Thus, *SETD2* loss may represent a hallmark of cancer, through the induction of genomic instability to an extent that is sufficient to generate a broad array of genetic defects, promote cancer evolution and promote the acquisition of further deleterious events that have metastatic and lethal consequences. A 2017 study reported that SETD2 regulates H4K16 acetylation through its modulation of H3K36 methylation[163]. As mentioned above, H4K16 acetylation is an important molecular determinant of the balance between HRR and NHEJ[157]. Thus, it is possible that the combination of *SETD2* loss together with *VHL* deletion potentiates the development of DNA double-strand repair breaks.

A number of studies have also evaluated the effects of *BAP1* loss on DNA repair. One study found that loss of *BAP1* sensitized ccRCC cell lines to radiation and PARP inhibitors[8]. Another study showed that BAP1 mediated poly (ADP-ribose)-dependent recruitment of the polycomb-repressive deubiquitinase complex (PR-DUB) to sites of DNA damage, which was phosphorylated by ATM, and subsequently promoted repair of DNA double-strand breaks[164]. As noted above, loss of *BAP1* destabilized MCRS1 and decreased mitotic stability[127]. These data suggest that BAP1 is regulator of DNA repair and helps to maintain copy number integrity, and its loss may foster genomic instability in ccRCC.

A few notable caveats must be noted when considering the evidence supporting a bona fide HRR defect in ccRCC. The first is that ccRCC is not typically sensitive to platinum and other DNA-damaging agents, which would be expected in a tumour with an HRR defect. The second is that although patterns associated with DNA mismatch repair have been identified in ccRCCs[141], this tumour type does not seem to possess a DNA damage pattern typical of those seen in tumours from patients with stereotypical germline defects in HRR genes. One possible explanation for this discrepancy is that the signature derived from HRR-typical *BRCA*-mutated tumours is sufficiently specific that it does not translate to other variants of HRR deficiency. Evidence supporting this possibility include a study that generated a homologous recombination defect (HRD) signature to identify tumours that have broad defects in HRR[165]. This HRD signature identifies tumours with vulnerabilities in HRR, despite not having mutations in classical HRR genes. Preliminary data suggest these signatures could help to identify whether a HRD exists in ccRCC and guide therapeutic approaches[166].

Drivers of lethality in ccRCC

One of the key questions in ccRCC cancer evolution is whether lethality is an innate characteristic or develops over time through the acquisition of additional genomic lesions. A number of studies over the past few years have demonstrated that both scenarios are operative, which may depend on the molecular signature of the ccRCC tumour in question[19,21].

ccRCC molecular subtypes

The TRACERx Renal study defined seven major clonal subtypes of ccRCC: VHL monodriver, PBRM1–SETD2, PBRM1–somatic copy number alterations (SCNAs), PBRM1–PI3K, VHL wild-type, multiple clonal driver and BAP1-driven[19]. These subtypes are genomically distinct and are prognostically heterogeneous. The TRACERx investigators assessed a number of determinants of lethality in these subtypes, including the wGII, which is a measure of the fraction of the tumour genome affected by SCNAs, the percentage of cells positive for the proliferation marker Ki67, the ITH index, the clonal structure and clinicopathological parameters[19].

Microhomology repair
An error-prone, alternative form of non-homologous end joining that involves alignment of microhomologous sequences internal to the broken ends before joining.

PARP inhibitors
A group of pharmacological inhibitors of the enzyme poly(ADP-ribose) polymerase (PARP).

Somatic copy number alterations
(SCNAs). Somatic changes to chromosome structure that result in gain or loss of copies of sections of DNA.

The VHL monodriver subcategory was largely found in small tumours, and demonstrated low ITH, wGII and Ki67 levels, and may therefore represent earlier tumours, or tumours that inherently lack the ability to develop additional lethal events. At the other end of the spectrum, the multiple clonal driver subtype showed high levels of wGII and Ki67 but relatively low levels of ITH, indicating that the simultaneous appearance of a small number of clonal drivers within this subtype is sufficient to create ccRCC clones with a strong selective advantage[19].

Prognostic implications of SCNAs

Several studies in addition to the TRACERx papers have shown that SCNAs are associated with poor clinical outcomes[11–14,167,168]. Partial or whole chromosomal losses or gains occur in chromosomes 5, 6, 7, 8, 9 and 14 at a relatively high frequency in ccRCC, and increase as a function of tumour size and grade[11,13].

Not only are higher overall SCNAs associated with more aggressive disease, specific chromosomal changes are linked to metastatic potential, suggesting a driver function. A 2009 study of 703 ccRCC tumours revealed that loss of chromosome 9p was associated with poor outcomes[12]. Subsequent analyses of chromosomal copy number changes in 111 patients with ccRCC revealed that the loss of chromosome 14q, together with a gain in chromosome 8q, was significantly associated with a poor prognosis[13,14]. In the TRACERx Renal study, 9p loss was identified as the factor most significantly associated with the formation of metastases[21]; 14q loss was significantly associated with development of metastasis prior to correction for multiple testing[21]. In addition, 9p loss in metastasizing clones was a significant predictor of poor overall survival after correction for known clinical variables[21]. Assessment of potential tumour suppressor or pro-oncogenic genes on these chromosomal regions identified several candidates. Chromosome 14q contains *HIF1A*, which appeared to act antagonistically to HIF2a in terms of its effects on ccRCC progression[169]. Chromosome 9p encodes *CDKN2A*, which generates p16 (also known as INK4A) — a critical G1 cell cycle regulatory protein. Chromosome 8q harbours the *MYC* gene, the upregulation of which is associated with aggressive tumour biology[170]. Copy-number alterations in these genes may be responsible for engendering a selective advantage for tumour cells; alternatively, the loss or gain of other genes on these chromosomes might cooperatively enhance cell fitness[21]. The next challenge is to understand the mechanisms underpinning SCNA, with the goal of either blocking its development, or targeting specific vulnerabilities in cells possessing these SCNAs.

Intratumoural heterogeneity

ITH is a major point of concern in the molecular characterization of ccRCC. Clinically relevant ITH is generated by mutations in key genes and by driver SCNAs[19]. The TRACERx researchers defined the ITH index as the number of subclonal drivers divided by the number of clonal drivers, where drivers included all driver mutations and driver SCNAs[19]. Future work will permit a more nuanced weighting of these driver events to better define tumour prognosis. Lower levels of ITH are seen at both ends of the tumour lethality spectrum. Tumours harbouring only *VHL* mutations as well as tumours with multiple driver mutations had low ITH. For tumours with multiple driver mutations, clonal genomic events probably confer an important selective advantage, permitting these cells to be the primary clone that generates the tumour mass and imparts aggressive tumour behaviour.

Genotypic divergence is likely limited by constraints in cellular viability and fitness, and may lead to phenotypic convergence and a manageable number of targetable pathways[171]. Further work is needed to define the phenotypes that are likely to be successfully targeted and identify tailored therapies that address these specific tumour features.

Sarcomatoid RCC

A special comment is needed to address sarcomatoid RCC, which is a particularly aggressive subset of ccRCC. Sarcomatoid variation is a histological designation assigned to highly aggressive tumours with underlying ccRCC features. A transcriptomic and genomic analysis of sarcomatoid RCC identified a significantly lower rate of chromosome 3p mutation overall, and an enrichment of *PTEN*, *TP53* and *RELN* mutations compared with ccRCC[110]. This pattern persisted regardless of whether tumours were sampled in areas harbouring sarcomatoid patterns or in regions with more traditional histology. Similar findings have been observed by others[172,173], and suggest that sarcomatoid ccRCC may be fundamentally different to non-sarcomatoid ccRCC in its early molecular pathogenesis. Interestingly, the overall mutational burden is broadly similar between sarcomatoid and non-sarcomatoid ccRCC[172,173], suggesting that mechanisms of mutation or a cell-type-specific threshold for mutational burden are similar in the two entities.

The tumour microenvironment

Tumour cells, and the diverse non-tumour cell types and structures found in immediate proximity to the tumour cells that make up the cancer 'organ', constitute the tumour microenvironment. These cells include endothelial cells, stromal cells and immune cells[174]. Multiple groups have assessed the diversity of intratumoural immune cells in ccRCC[113,120,175–180] (TABLE 1). One of the key questions is whether identifiable interactions exist between tumour genomic features and the microenvironment that might inform therapeutic decisions (FIG. 6).

One study that performed mass cytometric high-dimensional single-cell analysis of ccRCC tumour samples from 73 individuals with varying stages of disease[175] demonstrated diverse populations of immune cells and macrophage populations, with marked heterogeneity within and between samples from the same patient as well as between patients. Specific patterns of co-expressed T cells and macrophages were reported, including an association between the presence of exhausted and/or regulatory T cells and a specific subgroup of TAMs that expressed high levels of the MHC class II cell surface receptor HLA-DR, the monocytic

Table 1 | **Studies of immune cell populations in ccRCC**

Tissue	Study goal	Analytes and analytical tools	Key findings	Ref.
Primary ccRCC	Define ccRCC macrophage and T cell subtypes	Single cell mass cytometry; single cell RNA-seq	Described 17 tumour-associated macrophage and 22 T cell phenotypes. Identified prognostically relevant macrophage subcategories	175
Primary ccRCC; tumourgrafts	Characterize RCC TME	Bulk and single cell RNA-seq; IHC; WES	Identified novel immune/stromal transcripts. Defined an inflamed pan-RCC subtype	178
Primary ccRCC	Create clinically relevant ccRCC subtypes using RNA	RNA microarray; WES	Defined four biologically distinct clusters, with specific angiogenic and immune features. Linked subtypes to clinical outcome and treatment response	120
Primary ccRCC	Assess response to immunotherapy and identify genomic and microenvironmental response determinants	Bulk RNA-seq; WES	Combined immunotherapy plus anti-VEGF therapy showed trend towards improved outcome versus TKI. Response to immunotherapy higher in patients whose tumour transcriptomes showed T cell infiltrates. Presence of myeloid population attenuated immunotherapy response; addition of anti-VEGF therapy rescued immunotherapy effect	180
Primary ccRCC	Link genomic status to immune response	WES	PBRM1 deficiency was associated with improved response to immunotherapy	179
Primary ccRCC	Proteomic and phosphoproteomic data to subclassify ccRCC	Bulk RNA-seq; WES; reversed phase liquid chromatography; immobilized metal affinity chromatography	Identified genomically unstable ccRCC subtypes. Mapped metabolic changes in Krebs cycle and OXPHOS pathways. Defined $CD8^+$ inflamed, $CD8^-$ inflamed, metabolic immune desert and VEGF immune desert subtypes	119
Primary ccRCC. Isogenic Renca cell lines	Assess impact of *PBRM1* loss in immune response	WES; RNA-seq; multispectral IHC	*PBRM1*-null tumours were less immunogenic than other ccRCC tumours in a Renca murine model and in human tissue. PBRM1 deficiency reduced *IFGR2* transcription	113

ccRCC; clear cell renal cell carcinoma; IHC, immunohistochemistry; OXPHOS, oxidative phosphorylation; RCC, renal cell carcinoma; RNA-seq, RNA sequencing; TKI, tyrosine kinase inhibitor; TME, tumour microenvironment; VEGF, vascular endothelial growth factor; WES, whole exome sequencing.

markers CD38, CD64 and CD68, as well as the M2 polarization marker CD204. Other TAM subgroups were also identified that co-expressed protumoural markers such as CD163, CD204 and CD206, together with antitumoural TAM markers such as CD169. These 'subgroups were associated with more advanced tumours (that is, higher than stage 1); however, the presence of TAM subgroups that were not specifically associated with T cells was associated with worse progression-free survival than the presence of TAM subgroups associated with T cells. This study illustrates the tremendous diversity of T cell and TAM populations, and the challenges in identifying driver populations and therapeutic strategies. This study did not link these cell populations to tumour genomic features. Doing so will help our understanding of the interplay between tumour cell-specific features and the immune microenvironment.

Another study that compared the transcriptomes of uninvolved kidney tissue, bulk tumour tissue from patients with ccRCC and other RCC tumour subtypes and tumourgrafts from patient-derived xenografts identified a set of genes expressed by immune and stromal cells that were specific to ccRCC, for which they coined the term empirically defined gene signature of the tumour microenvironment (eTME). Of these eTME genes, 2,080 were expressed 3-fold higher and 904 were expressed 20-fold or higher in the immune and/or stroma component than in the tumour[178]. Of note, substantial proportions of immune signature genes listed in other data sets (that is, 65%, 22% and 9% of immune-associated genes from the pan-tumour transcriptomic immune profiles represented in the Immunome[181], ESTIMATE[182] and Winslow[183] data sets, respectively) were not abundantly expressed in the eTME gene set. Comparison of the eTME gene set with TCGA RCC data sets, including those representing ccRCC, papillary and chromophobe RCC led to the identification of two RCC subgroups — one that was associated with high levels of transcripts encoding immune and complement components (considered to be highly inflamed), and a non-inflamed group that was enriched for markers of angiogenesis, plasmacytoid dendritic cells and mast cells. The prevalence of inflamed tumours in patients with type I papillary RCC was low. Inflamed ccRCC-type tumours were enriched for *BAP1* mutations. Among a single-centre cohort, patients with inflamed ccRCC tumours manifested indicators of poor prognosis as described by the International Metastatic RCC Database Consortium, including anaemia and thrombocytosis[184].

A 2019 study that assessed the immune microenvironment in 409 patients with ccRCC who had been treated with sunitinib or pazopanib by unsupervised clustering of microarray data identified four molecularly distinct clusters, with varying treatment response rates and overall patient survival[120]. For example, Cluster 4, which was associated with worse overall survival and poor International Metastatic RCC Database Consortium risk categorization, was characterized by elevated levels of the checkpoint protein PD-L1 and mutations in *BAP1* and *TP53*, whereas mutations in *PBRM1* were infrequent. Tumours harbouring *PBRM1* mutations were associated with a higher expression of genes involved in angiogenesis, whereas tumours harbouring *BAP1* mutations were associated with lower expression of angiogenesis genes. Pathway analysis demonstrated elevated inflammatory signatures,

including IFNγ response in Cluster 4, and the highest immune score using the ESTIMATE[182] algorithm, which uses gene expression data to output the estimated levels of infiltrating stromal and immune cells and estimated tumour purity. Higher macrophage infiltration was associated with worse response to antiangiogenic therapy and with poor overall survival. Scoring based on angiogenesis (Angio) and macrophage (Macrophage) status demonstrated that patients who had Angio^{lo}Macrophage^{hi} tumours demonstrated poorer outcomes in response to antiangiogenic therapy than those with Angio^{hi}Macrophage^{lo} tumours.

Another study of the clinical outcomes of 301 patients with advanced ccRCC treated with the PD-L1 inhibitor atezolizumab, the combination of atezolizumab plus the anti-VEGF antibody bevacizumab or sunitinib reported that patients could be grouped into clinically relevant categories based on their angiogenesis, effector T cell, and myeloid cell signatures[180]. Patients in the angiogenesis 'high' subgroup were more likely than those in the other two subgroups to respond to sunitinib therapy, whereas those in the effector T cell 'high' group were more likely to respond to the combination of atezolizumab plus bevacizumab. Further division of the subgroups demonstrated that the absence of a myeloid population was associated with a trend towards improved response to atezolizumab monotherapy relative to sunitinib; this trend was reversed for tumours associated with a myeloid population[185]. Additional analyses demonstrated that tumours with *PBRM1* mutations were significantly associated with response to sunitinib, but not with response to either atezolizumab or the combination of atezolizumab plus bevacizumab, which is in line with other studies showing that tumours with *PBRM1* mutations are associated with a higher expression of genes involved in angiogenesis[185].

A 2019 multiparametric analysis of 103 treatment-naive ccRCC and 84 matched uninvolved tissue samples[119] led to the identification of a subset of ccRCC characterized by chromosome-level genomic instability and very poor prognosis. Proteomic analyses of this tumour subset demonstrated an upregulation of components involved in glycolysis and a downregulation of those involved in oxidative phosphorylation relative to levels in normal tissue; however, the changes in oxidative phosphorylation components were observed at a protein but not at the transcriptome level, with late-stage tumours showing increased oxidative phosphorylation compared with earlier stage tumours. Classification of samples according to their immune cell signatures divided tumours into CD8$^+$-inflamed tumours, characterized by a high degree of CD8$^+$ T cell infiltration; CD8$^-$-inflamed tumours, characterized by the presence of innate immune cells such as dendritic cells and macrophages; metabolic immune desert tumours, characterized by low levels of immune cells and a unique metabolic profile defined by increased expression of proteins involved in oxidative phosphorylation and glycolysis; and VEGF immune desert tumours, characterized by an enrichment of endothelial cells and angiogenesis. Notably, CD8$^+$-inflamed tumours were enriched for *BAP1* mutations, whereas the VEGF immune desert tumours had a higher proportion of *PBRM1* mutations.

More recently, transcriptomic mapping in PBRM1-null mice showed that immune gene expression in the murine tumours closely paralleled that seen in the

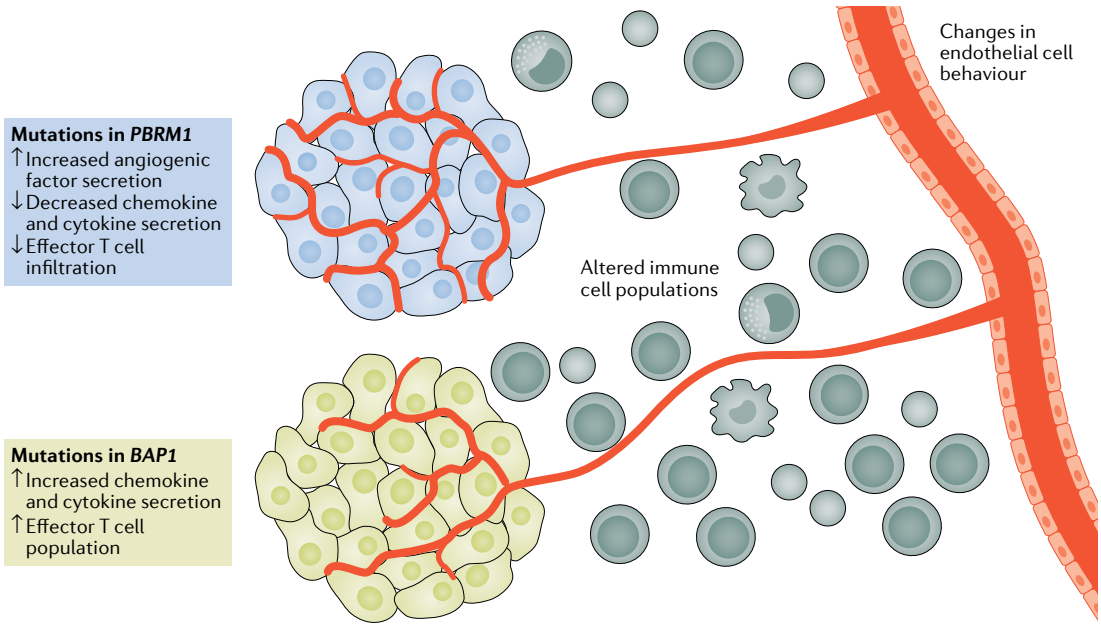

Fig. 6 | **Alterations in the immune microenvironment in clear cell renal cell carcinoma.** Tumour cell genomic features might affect the expression of cell surface receptors, as well as the synthesis and secretion of immunomodulatory chemokines, cytokines and proangiogenic factors. For example, *PBRM1* mutations increase angiogenesis and decrease chemokine and cytokine production, whereas tumours bearing *BAP1* mutations are associated with an inflamed microenvironment.

ccRCC TCGA data set[113]. In both animal models and in human samples, loss of *PBRM1* was associated with low levels of immune cell infiltration, and *Pbrm1*-null tumours were more resistant than *Pbrm1*-intact tumours to treatment with an anti-PD-1 antibody. Moreover, loss of *PBRM1* attenuated IFNγ-mediated transcription of chemoattractant chemokines — an effect that was at least partially due to loss of PBRM1-mediated upregulation of interferon-γ receptor 2 (*IFNGR2*) gene expression. In contrast to that study, however, a separate analysis of 63 patients with metastatic ccRCC treated with checkpoint-blocking antibodies indicated that *PBRM1*-mutated tumours were more likely than other tumours to respond to checkpoint antibody therapy[179]. A follow-up study by the same group demonstrated similar outcomes[186]. The contradictory results of these studies may be due to patient selection or to other confounding factors including prior therapies.

Treatment implications

Treatment of ccRCC is based largely on clinical parameters, despite the clear existence of molecular heterogeneity in these tumours. A long-term goal is to incorporate the molecular information that can be obtained from ccRCC into existing treatment algorithms. The following section outlines some potential opportunities to achieve this goal.

Prevention and early detection

Despite our understanding that chromosome 3p loss is an early and important driver of ccRCC ontogeny, the mechanism underlying its loss in ccRCC precursor cells has not yet been formally defined. Improved understanding of the aetiology of 3p loss could lead to the identification of strategies to prevent ccRCC development, particularly in high-risk populations, such as in individuals with germline losses in the *VHL* gene.

The early detection of 3p loss is also desirable and could in principle be achieved through blood- or urine-based detection systems with sufficient sensitivity to detect small amounts of genomic material harbouring 3p loss. For example, circulating tumour DNA (ctDNA) assays have the potential to capture specific SCNAs and identify 3p loss. The successful development of such an assay would encourage the development of treatment strategies to selectively target cells harbouring 3p loss, thereby eradicating the foundational clones responsible for ccRCC development.

Identification of lethal primary tumours

The ability to identify patients with non-metastatic ccRCC who are at a high risk of developing metastatic disease may aid the development of targeted adjuvant therapy strategies to prevent metastasis through the ability to enrich clinical trials with individuals at greatest need of adjuvant therapy, and ideally, by targeting the molecular drivers of increased risk. A number of nomograms and algorithms have been generated to stratify patients according to their risk of metastasis on the basis of clinicopathological data[187–189]. However, more detailed understanding of the impact of tumour genomic features on disease outcomes may help refine these algorithms. As an example, the seven TRACERx groups can be divided into three broad patterns of metastasis evolution: linear evolution, including patients with VHL monodriver disease; branched evolution, including patients with PBRM1–SCNA, PBRM1–SETD2 and PBRM1–PI3K tumours; and punctuated evolution, including those with BAP-driven, multiple clonal driver and VHL wild-type ccRCC[21]. Tumours with linear evolution patterns were less likely than tumours in the other groups to metastasize, and more likely to be successfully treated by nephrectomy without the need for adjuvant therapy. Patients in the branched evolution group tended to have an intermediate prognosis, as surgical removal of the primary tumour in these patients reduced the pool of potential metastatic subclones. Tumours with punctuated evolution patterns showed evidence of multiple clonal drivers and were considered to be at a high risk of metastasis and disease progression. These individuals could be considered for adjuvant therapy trials, and if discovered with synchronous metastases, are most likely to benefit from first-line systemic therapy as opposed to cytoreductive nephrectomy.

The continued development and validation of molecular subclassifiers will further improve prognostication of ccRCC. As these tumour subtypes become more robustly defined, ctDNA could increasingly be used to provide an aggregate readout of tumour genomic features to identify clones associated with tumour aggressivity and metastatic potential[190,191]. To date, very few studies have assessed the potential utility of ctDNA assays in ccRCC[192]. Despite its clear potential use, a number of questions remain with regard to the best approach to developing clinically impactful ctDNA platforms[193] and additional work is therefore required to bring this approach into routine practice for patients with ccRCC.

Targeting genomic vulnerabilities

The development of a molecularly defined taxonomy of ccRCC provides a parallel opportunity to develop subtype-tailored therapeutic options (TABLE 2). At this point in time, ccRCC treatments have not reached the granularity achieved for other tumour types such as breast and lung carcinomas[194,195]. Nonetheless, an understanding of ccRCC biology has already yielded considerable advances in therapies, and further work over the coming years will undoubtedly continue this trend.

The near-obligate loss of *VHL* in ccRCC has spurred efforts to develop targeting strategies with synthetic lethality[196] and to rescue the function of point-mutated VHL protein[75,197,198]. The metabolic consequences of *VHL* loss have been elegantly defined[199] and this research has led to the testing of glutaminase inhibitors in ccRCC[200,201]. Upregulated HIF[35] also represents a target for ccRCC, with HIF2α inhibitors showing promising results in preclinical and clinical studies[202,203]. Ongoing trials are testing HIF2α inhibitors in combination with tyrosine kinase inhibitor (TKIs) in advanced ccRCC[204] and as monotherapy in patients with hereditary VHL disease[205]. Preliminary data from these studies show that the small molecule HIF2α inhibitor MK6482 is active both in advanced ccRCC[206] and in ccRCC associated with VHL disease[207].

Circulating tumour DNA
(ctDNA). DNA in the bloodstream that comes from cancerous cells and tumours.

Synthetic lethality
The concept whereby the simultaneous mutation of two genes leads to the lethality of a cell, but the mutation of one alone is viable.

Table 2 | **Therapeutic opportunities to target specific tumour features**

Tumour alteration	Therapeutic opportunities	Knowledge gap
VHL and/or 3p loss	Prevention; early detection	Initiating factors for 3p loss; sensitive assays for early clonal evolution
Mutations in chromatin remodelling genes	Synthetic lethal strategies	Mechanism(s) of oncogenesis
Upregulation of oncogenic pathways	Targeted agents	Key drivers of pathway upregulation and/or mechanisms of resistance
Modulation of the immune microenvironment	Tailored intervention(s) to modulate the immune response	Mechanism(s) of microenvironmental alteration(s) and their association with tumour genotype and phenotype

The mechanistic consequences of *BAP1*, *SETD2* and *PBRM1* mutations are another area of active investigation. Although prognostic differences exist between tumours that harbour these mutations, distinct and targetable molecular features of these subtypes remain to be determined.

Loss of *BAP1* was shown to increase levels of the histone methyltransferase enhancer of zeste homologue 2 (EZH2) in a mesothelioma model[208], and increased sensitivity of *BAP1*$^{-/-}$ cells to EZH2 inhibition. However, contradictory findings in *BAP1*-null uveal melanoma suggest that the observations made in mesothelioma cannot be generalized to all cancer types associated with *BAP1* mutations[209]. An EZH2 inhibitor, tazemetostat, is currently in clinical trials. At this point in time, the applicability of this approach to ccRCC harbouring *BAP1* mutations is not known.

Preclinical studies demonstrated that deficiency of SETD2 increases the sensitivity of cultured cells to inhibitors of the cell cycle regulator WEE1. This effect was mediated via the decreased expression and increased degradation of the ribonuclease reductase subunit, RRM2, resulting in nucleotide depletion, replication stress and cell death[210]. The WEE1 inhibitor adavosertib demonstrated preclinical efficacy in H3K36me3-deficient tumour xenografts[210]. A number of clinical trials of adavosertib are now underway[140], including a study focused on SETD2-deficient tumours. In addition, a preclinical study demonstrated that synthetic lethality exists between SETD2 deficiency and PI3Kβ inhibition in a ccRCC model, and could represent another potential therapeutic approach[211].

Agents that target TORC1 signalling are approved for treatment of patients with advanced ccRCC, but have shown modest efficacy[212,213], as have agents that target AKT[214]. In the past few years, the PI3K pathway has been shown to modulate elements of HRR; moreover, inhibition of PI3K pathway signalling might sensitize various tumour subtypes to PARP inhibition[215,216]. In line with this finding, the RCC 786-0 cell line, which has elevated PI3K pathway signalling and is relatively resistant to PARP inhibition, could be sensitized to PARP inhibitors through concomitant blockade of PI3K pathway signalling[165]. These observations provide a potential new direction with which to modulate ccRCC signalling to render the tumour more susceptible to DNA repair-targeted therapy.

Targeting the tumour microenvironment

Improved understanding of the pro-angiogenic consequences of *VHL* mutation in ccRCC spawned a panoply of agents that targeted either VEGF[217,218] or its receptors[219–224] in ccRCC. These agents moved the treatment of ccRCC from the use of relatively toxic and inconsistently effective cytokine-based therapy to the predominant use of orally bioavailable agents with consistent efficacy and manageable adverse effects. Use of these agents in patients with hereditary VHL disease resulted in a salutary effect in those harbouring ccRCC[225,226], and had some benefit in other affected organs, including in pancreatic and central nervous system lesions.

As the various phenotypic and genotypic subvariants of ccRCC become increasingly well defined, it will be essential to understand how the molecular features of specific subvariants influence the tumour-immune microenvironment and how this information can be used to add precision to therapy selection. As an example, in sarcomatoid RCC, PD-L1 is upregulated relative to levels observed in non-sarcomatoid ccRCC, with increased numbers of T cells in the tumour microenvironment[227,228]. Emerging data suggest that patients with sarcomatoid RCC demonstrate a superior response to checkpoint antibody therapy[229], with larger confirmatory analyses currently underway.

A number of studies in the past few years have assessed correlations between specific molecular signatures, the tumour microenvironment and clinical outcome. Multiple studies have shown that *PBRM1* mutations are associated with upregulated angiogenesis and downregulated immune cell infiltration. In agreement with the preclinical data described earlier[104,111], data from the RECORD-3 study indicated that patients with *PBRM1*-mutated tumours responded well to anti-angiogenic therapy[131]. Similarly, the IMmotion 150 study also demonstrated that patients with *PBRM1*-mutated tumours responded better to anti-angiogenic agents than those without *PBRM1* mutations. Moreover, this directionality was reversed with PD-L1 blocking therapy whereby patients with *PBRM1*-mutated tumours showed a trend towards worse outcome after atezolizumab treatment[180]. This finding is in contradiction to data reported by other researchers, which indicated that *PBRM1* mutations were associated with a better response to checkpoint antibody therapy than other tumour types[179,186]. Clearly, more work needs to be done to provide clarity on the impact of this commonly mutated gene on immune response in ccRCC.

As described earlier, B*AP1*-mutated tumours are associated with an inflamed immune microenvironment[119,178], suggesting that some form of immune-targeting strategy may benefit these patients. Less is known about the impact of *SETD2* mutations on the immune microenvironment and treatment response, with ongoing efforts aimed at better characterizing their effect.

cGAS–STING pathway
A component of the innate immune system that functions to detect the presence of cytosolic DNA and triggers the expression of inflammatory genes.

The expression of AXL on both ccRCC tumour cells[49] and on immune suppressive cell populations in the tumour microenvironment[230,231] provides additional opportunities for targeted, integrated therapeutic approaches. Clinical trials that aim to test the efficacy of combined targeting of TAM family members in combination with immunotherapy are underway, using a combination of cabozantinib and nivolumab[232].

Targeting tumour heterogeneity

Tumour heterogeneity might hinder our ability to fully characterize the genomic drivers of ccRCC[19], and may complicate the choice of molecularly targeted agent against key driver mutations. If sufficiently sensitive, ctDNA may help to identify key drivers of tumour biology and therapy selection.

Similarly, tumour heterogeneity may decrease the impact of targeted therapy in the treatment of ccRCC, especially if the molecules targeted are downstream of the driver events. By targeting the drivers of heterogeneity, it may be possible to decrease or prevent further clonal and subclonal evolution; however, the very presence of this aberrant programming might confer specific vulnerabilities to the tumour cell.

Some evidence suggests that the tumour response to immunotherapy may be enhanced by tumour heterogeneity[171]. The diverse repertoire of tumour neoantigens might enhance the immune response to checkpoint inhibitors by providing a larger number of opportunities for cytotoxic T cells to interact with and kill tumour cells[233]. Tumours that have high levels of microsatellite instability (so-called MSI-high tumours) are particularly responsive to checkpoint inhibitor therapy[234,235]. Despite the intermediate level of tumour heterogeneity observed in ccRCC[141], this tumour type is quite responsive to checkpoint inhibitory therapy[236,237], and this response seems unrelated to the qualitative characteristics of the ccRCC mutational repertoire, including insertions and deletions, and associated frameshift mutational burden[180]. The reason why ccRCCs respond well to immunotherapy has not been fully elucidated, and future research investigating the cGAS–STING pathway[238], which enhances a type I interferon response and increases secretion of immunostimulatory chemokines and cytokines, as well as other mechanisms of immunogenicity, may provide clues.

Conclusions

In the past decade, our understanding of the determinants of ccRCC ontogeny and the drivers of heterogeneity and lethality has reached the point where we can now begin to describe the cogent mechanistic pathways that control tumour ontogeny and progression, and create clinically relevant subcategories of ccRCC that may benefit from subtype-specific therapeutic interventions. Ongoing work to elucidate the impact of driver gene mutations on ccRCC biology will help to create the necessary framework to begin developing more targeted therapeutic approaches for patients with ccRCC. As our understanding of the molecular biology of ccRCC expands, a parallel evolution of diagnostic, imaging and therapeutic tools will enable us to translate these advances into substantive improvements in the prevention, early detection and treatment of ccRCC.

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